

CSE260

Assignment 02

This assignment must be hand-written. Show ALL steps in ALL questions.

Marks: 25

Q1: 5+5= 10

Q2: 5

Q3: 5+5=10

1. Simplify the following boolean expressions to minimum number of literals:

a. $(A + B)(A+B)(A+C)$
 $= AA + AB + AB + BB (A+C)$
 $= A + AB + B (A+C)$
 $= A + AB + AB + BC$
 $= A + AB + BC$
 $= A (1+B) + BC$
 $= A + BC$

b. $(A'B+B')BCC'+CB'C+D$
 $= A'BBCC' + B'BCC' + CB'C + D$
 $= 0 + 0 + B'C + D$
 $= B'C + D$

2. Find the complement of the following expression:

$$(x' + y + z')(x + y')(x' + z)$$
$$= (x'.y.z') + (x.y') + (x'.z)$$
$$= (x.y'.z) + (x'.y) + (x.z')$$

3. Find out SOP and POS for the following [You cannot use truth table to solve these]:

a. $F(A,B,C,D,E) = A + B'CD'$

$$= \sum(4, 5, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$$
$$\prod(0, 1, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

$$= \sum(2, 3, 10, 11, 12, 13, 14, 15, 18, 19, 26, 27, 28, 29, 30, 31)$$
$$= \prod(0, 1, 4, 5, 6, 7, 8, 9, 16, 17, 20, 21, 22, 23, 24, 25)$$

SOP

$$\begin{aligned}
F(A,B,C,D,E) &= A + B'CD' \\
&= A(B+B')(C+C')(D+D')(E+E') + B'CD'(A+A')(E+E') \\
&= (AB + AB')(CD+CD'+C'D+C'D')(E+E') + B'CD'(AE+AE'+A'E+A'E') \\
&= (AB + AB')(CDE+CD'E+C'DE+C'D'E + CDE'+CD'E'+C'DE'+C'D'E') + AEB'CD'+AE'B'CD'+A'EB'CD'+A'E'B'CD' \\
&= ABCDE+ABCD'E+ABC'DE+ABC'D'E + ABCDE'+ABCD'E'+ABC'DE'+ABC'D'E' + AB'CDE+AB'CD'E+AB'C'DE+AB'C'D'E + AB'CDE'+AB'CD'E'+AB'C'DE'+AB'C'D'E'+ AB'CD'E+AB'CD'E'+A'B'CD'E+A'B'CD'E' \\
&= \sum(11111, 11101, 11011, 11001, 11110, 11100, 11010, 11000, 10111, 10101, 10011, 10001, 10110, 10100, 10010, 10000, 10101, 10100, 00101, 00100) \\
&= \sum(00100, 00101, 10000, 10001, 10010, 10011, 10100, 10101, 10110, 10111, 11000, 11001, 11010, 11011, 11100, 11101, 11110, 11111) \\
&= \sum(4, 5, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)
\end{aligned}$$

POS

$$\begin{aligned}
F(A,B,C,D,E) &= A + B'CD' \\
&= (A+B')(A+CD') \\
&= (A+B')(A+C)(A+D') \\
&= (A+B'+CC')(A+C+BB')(A+D'+BB') \\
&= (A+B'+C)(A+B'+C')(A+C+B)(A+C+B')(A+D'+B)(A+D'+B') \\
&= (A+B'+C+DD')(A+B'+C'+DD')(A+C+B+DD')(A+C+B'+DD') \\
&\quad (A+D'+B+CC')(A+D'+B'+CC') \\
&= (A+B'+C+D)(A+B'+C+D')(A+B'+C'+D)(A+B'+C'+D') \\
&\quad (A+C+B+D)(A+C+B+D')(A+C+B'+D)(A+C+B'+D') \\
&\quad (A+D'+B+C)(A+D'+B+C')(A+D'+B'+C)(A+D'+B'+C') \\
&= (A+B'+C+D+EE')(A+B'+C+D'+EE')(A+B'+C'+D+EE')(A+B'+C'+D'+EE') \\
&\quad (A+C+B+D+EE')(A+C+B+D'+EE')(A+C+B'+D+EE')(A+C+B'+D'+EE') \\
&\quad (A+D'+B+C+EE')(A+D'+B+C'+EE')(A+D'+B'+C+EE')(A+D'+B'+C'+EE') \\
&= (A+B'+C+D+E)(A+B'+C+D+E')(A+B'+C+D'+E)(A+B'+C+D'+E')(A+B'+C'+D+E)(A+B'+C'+D+E') \\
&\quad (A+B'+C'+D'+E)(A+B'+C'+D'+E')(A+B'+C'+D'+E')(A+B'+C'+D'+E')(A+B'+C'+D'+E')(A+B'+C'+D'+E') \\
&\quad (A+B'+C'+D'+E)(A+B'+C'+D'+E')(A+B'+C'+D'+E)(A+B'+C'+D'+E)(A+B'+C'+D'+E)(A+B'+C'+D'+E) \\
&\quad (A+B'+C'+D'+E)(A+B'+C'+D'+E)(A+B'+C'+D'+E)(A+B'+C'+D'+E)(A+B'+C'+D'+E)(A+B'+C'+D'+E) \\
&= \prod(01000, 01001, 01010, 01011, 01100, 01101, 01110, 01111, 00000, 00001, 00010, 00011, 01000, 01001, 01010, 01011, 00010, 00011, 00110, 00111, 01010, 01011, 01110, 01111) \\
&= \prod(8, 9, 10, 11, 12, 13, 14, 15, 0, 1, 2, 3, 8, 9, 10, 11, 2, 3, 6, 7, 10, 11, 14, 15) \\
&= \prod(0, 1, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)
\end{aligned}$$

$$b. \quad F(V,W,X,Y,Z) = WY + WX + X'Y$$

SOP

$$F(V,W,X,Y,Z) = WY + WX + X'Y$$

$$= WY (V+V')(X+X')(Z+Z') + WX (V+V')(Y+Y')(Z+Z') + X'Y (V+V')(W+W')(Z+Z')$$

$$= WY (VX+VX'+V'X+V'X')(Z+Z') + WX (VY+VY'+V'Y+V'Y')(Z+Z') + X'Y (VW+VW'+V'W+V'W')(Z+Z')$$

$$= WY (VXZ+VXZ'+VX'Z+VX'Z'+V'XZ+V'XZ'+V'X'Z+V'X'Z') + WX (VYZ+VYZ'+VY'Z+VY'Z'+V'YZ+V'YZ'+V'Y'Z+V'Y'Z') + X'Y (VWZ+VWZ'+VW'Z+VW'Z'+V'WZ+V'WZ'+V'W'Z+V'W'Z')$$

$$= WYVXZ + WYVXZ' + WYVX'Z + WYVX'Z' + WYV'XZ + WYV'XZ' + WYV'X'Z + WYV'X'Z' + WXVYZ + WXVYZ' + WXVY'Z + WXVY'Z' + WXV'YZ + WXV'YZ' + WXV'Y'Z + WXV'Y'Z' + X'YVWZ + X'YVWZ' + X'YVW'Z + X'YVW'Z' + X'YV'WZ + X'YV'WZ' + X'YV'W'Z + X'YV'W'Z'$$

$$= VWXYZ + VWXYZ' + VWX'YZ + VWX'YZ' + V'WXYZ + V'WXYZ' + V'WX'YZ + V'WX'YZ' + VWXYZ + VWXYZ' + VWXY'Z + VWXY'Z' + V'WXYZ + V'WXYZ' + V'WX'YZ + V'WX'YZ' + VWX'YZ + VWX'YZ' + VW'X'YZ + VW'X'YZ' + V'WX'YZ + V'WX'YZ' + V'W'X'YZ + V'W'X'YZ'$$

$$= \sum(11111, 11110, 11011, 11010, 01111, 01110, 01011, 01010, 11111, 11110, 11101, 11100, 01111, 01110, 01101, 01100, 11011, 11010, 10011, 10010, 01011, 01010, 00011, 00010)$$

$$= \sum(31, 30, 27, 26, 15, 14, 11, 10, 31, 30, 29, 28, 15, 14, 13, 12, 27, 26, 19, 18, 11, 10, 3, 2)$$

$$= \sum(2, 3, 10, 11, 12, 13, 14, 15, 18, 19, 26, 27, 28, 29, 30, 31)$$

POS

$$\begin{aligned}
F(V,W,X,Y,Z) &= WY + WX + X'Y \\
&= (WY + WX + X')(WY + WX + Y) \\
&= (WY + (W + X')(X + X'))(WY + (W + Y)(X + Y)) \\
&= (WY + W + X')(WY + (W + Y)(X + Y)) \\
&= (W + X' + W)(W + X' + Y)((W + Y)(X + Y) + W)(W + Y)(X + Y) + Y \\
&= (W + X')(W + X' + Y)(W + Y)(W + X + Y)(X + Y) \\
&= (W + X' + VV')(W + X' + Y + VV')(W + Y + VV')(W + X + Y + VV')(X + Y + VV') \\
&= (V + W + X')(V' + W + X')(V + W + X' + Y)(V' + W + X' + Y)(V + W + Y)(V' + W + Y)(V + W + X + Y)(V' + W + X + Y)(V + X + Y)(V' + X + Y) \\
&= \\
&(V + W + X' + Y)(V + W + X' + Y')(V' + W + X' + Y)(V' + W + X' + Y')(V + W + X' + Y + Z)(V + W + X' + Y + Z')(V' + W + X' + Y + Z)(V' + W + X' + Y + Z')(V + W + X + Y)(V + W + X' + Y)(V' + W + X + Y)(V' + W + X' + Y)(V + W + X + Y + Z)(V + W + X + Y + Z')(V' + W + X + Y + Z)(V' + W + X + Y + Z')(V + W + X + Y)(V + W' + X + Y)(V' + W + X + Y)(V' + W' + X + Y) \\
&= \\
&(V + W + X' + Y + Z)(V + W + X' + Y + Z')(V + W + X' + Y' + Z)(V + W + X' + Y' + Z')(V' + W + X' + Y + Z)(V' + W + X' + Y + Z')(V' + W + X' + Y' + Z)(V' + W + X' + Y' + Z')(V + W + X' + Y + Z)(V + W + X' + Y + Z')(V' + W + X' + Y + Z)(V' + W + X' + Y + Z')(V + W + X + Y + Z)(V + W + X + Y + Z')(V' + W + X + Y + Z)(V' + W + X + Y + Z')(V + W + X + Y + Z)(V + W + X' + Y + Z)(V + W + X + Y + Z')(V' + W + X + Y + Z)(V' + W + X + Y + Z')(V + W + X + Y + Z)(V + W' + X + Y + Z)(V + W' + X + Y + Z')(V' + W + X + Y + Z)(V' + W + X + Y + Z')(V' + W' + X + Y + Z)(V' + W' + X + Y + Z') \\
&= \prod (00100, 00101, 00110, 00111, \\
&\quad 10100, 10101, 10110, 10111, 00100, 00101, 10100, 10101, 00000, 00001, 00100, 00101, 10000, 10001, 10100, 10101, 00000, 00001, 10000, 10001, 00000, 00001, 01000, 01001, 10000, 10001, 11000, 11001) \\
&= \prod (4, 5, 6, 7, 20, 21, 22, 23, 4, 5, 20, 21, 0, 1, 16, 17, 1, 0, 4, 5, 16, 17, 20, 21, 0, 1, 8, 9, 16, 17, 24, 25) \\
&= \prod (0, 1, 4, 5, 6, 7, 8, 9, 16, 17, 20, 21, 22, 23, 24, 25)
\end{aligned}$$